

Dr.Babasaheb Ambedkar Technological University

Department of Electronics and Telecommunication Engineering

Semester Examination

Subject: CONTROL ENGINEERING (BET 601)

semester :VI

Date:

Time:

Max.Marks : 70

Instruction to Students

1. Assume suitable data wherever necessary & state it clearly.
 2. Q.1 is compulsory and carry 10 Marks whereas Q.2 to Q.7 carries 12 Marks each.
 3. Attempt any five question from Q.2 to Q.7.
 4. Necessary data is given in respective question. If such data is not given, it means that the knowledge of that part is Examination.
- If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly.

10M

- 1) The term 'reset control' refers to
 - a) Sensitivity
 - b) Non – linearity
 - c) Disturbance
 - d) All of these
- 2) When the gain K of a system is increased , the steady state error of the system
 - a) Increases
 - b) Remains same
 - c) Decreases
 - d) None of these
- 3) The term 'reset control' refers to
 - a) Proportional
 - b) Integral
 - c) Derivative
 - d) PID
- 4) The roots of Characteristics equation lie on imaginary axis the system is
 - a) Stable
 - b) Unstable
 - c) Conditionally stable
 - d) Marginally stable
- 5) Equivalent Analogy for torque in Mechanical translational system is
 - a) Force
 - b) Man
 - c) Acceleration
 - d) Displacement
- 6) In the signal flow graph ,the Branches represents
 - a) variable
 - b) Gain
 - c) Parameters
 - d) All of these
- 7) There are 3 Zeros and 2 poles of GH(S).There will be ____ Loci.
 - a) Three
 - b) Two
 - c) Five
 - d) One
- 8) Gain crossover frequency at which magnitude is
 - a) Zero
 - b) Unity
 - c) Infinity
 - d) Constant
- 9) Phase crossover frequency at which phase is
 - a) 90
 - b) 180
 - c) -180
 - d) 0
- 10) Gain margin is the reciprocal of

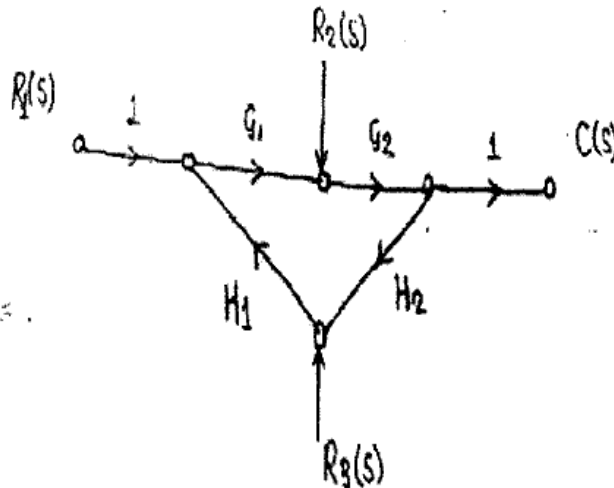
- a) Magnitude b) Phase c) Both a& b d) None of these

Q.2 A) Design a transfer function model for the Translational Mechanical System.

5M

B) Obtain Overall Transfer Function for SFG given below.

7M



Q.3 A) Determine the Stability for the $S^4 + 2S^2 + 1 = 0$

6M

B) Explain Basic Characteristics of Feedback Control System

6M

Q.4 A) For the System having <https://www.batuonline.com>

5M

$$\frac{C(S)}{R(S)} = \frac{40(S + 2)}{S(S + 1)(S + 4)}$$

Determine

i) Type of System iii) All Error Coefficient iii) ess for ramp input with Magnitude 4

B) Define

4M

i) Delay Time

iii) Peak Time

ii) Rise Time

iv) Settling Time

C) Find Natural frequency, Maximum overshoot and Settling time for system having

3M

$$\frac{C(S)}{R(S)} = \frac{8}{S^2 + 4S + 8}$$

Q.5 A) Sketch the Bode Plot for the transfer function.

8M

$$\frac{1000(0.02S + 1)}{(S + 1)(0.1S + 1)(0.01S + 1)}$$

Determine W_{gc} , W_{pc} , G.M., P.M.

- B) Draw the Nyquist Plot for the System and state stability for

4M

$$G(S)H(S) = \frac{100}{S^2(S+2)(S+4)}$$

- Q.6 A) Find state transition Matrix of system define by following state model also find transfer function of the system.

6M

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -5 & -25 & -5.5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ 25 \\ -120 \end{bmatrix} \quad \text{and } y = [1 \ 0 \ 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

- B) Evaluate Controllability of following system

6M

$$A = \begin{bmatrix} -2 & 0 \\ 0 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \quad C^T = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{And}$$

Evaluate Observability of following system

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}, \quad C = \begin{bmatrix} 3 & 0 & 0 \\ 4 & 0 & 0 \end{bmatrix}$$

- Q.7 A) The OLTF of Unity feedback system is given by,

12M

$$G(S) = \frac{K}{(S+3)(S+5)(S^2+2S+2)}$$

Draw the complete Root Locus diagram for this system,

-----BEST OF LUCK!-----